

wellmodel.app — v3.4 PBU rebuild: known limitations

Summary of the rebuild

The shut-in module was rebuilt from a lumped single-pressure gas-tank formulation to a cell-by-cell compressible hydrostatic column solver. Key physics now captured rigorously:

- **Distributed P, T, ρ_g per cell** — no more single averaged-pressure or averaged-temperature shortcut for a 12,000 ft well
- **Exact mass conservation** — reservoir afterflow adds lb-mass via $\rho_{sc} \times q_{res} \times dt$, with no Z in the denominator anywhere (previous formulation had a Z double-count that made PBU ~18% too fast)
- **Self-consistent hydrostatic profile** — verified to 0.0002 psi max error across all 20 cells in the probe runs
- **Coupling to radial reservoir grid** — BHP is bounded by the near-well cell pressure $P_{cell}[0]$, which itself evolves on the correct diffusion time scale $\eta = k/(\phi\mu c_t)$

The full recovery curve shape matches field intuition:

- WBS-dominated in the first minute
- Near-wellbore radial complete by ~30 min (BHP catches up to $P_{cell}[0]$)
- Far-field drainage on r_e^2/η time scale (days to weeks)
- 90-day recovery reaches ~97% of Pr for the screenshot parameters

Known limitation: one-step transient at shut-in

Observation: At $t = 0$ of shut-in, BHP briefly dips below its flowing value (by 300–500 psi for a deep gas well flowing at high rate), then recovers over the next ~30 seconds of simulated time.

Root cause: The flowing-state pressure cascade (Step D, else-branch in stepTransient) does a downward cascade from BHP to WHP, then overrides $P[0]$ with $\max(P_{sep}+3, P[0])$ as a bottom-hole-to-surface consistency check. This creates small inconsistencies in the middle cells — their $P[k]$ values are not fully mass-consistent with the flowing BHP. The new rigorous shut-in solver reads those $mG[k]$ values, totals them up, and finds the hydrostatic-

only BHP that matches that total. Result: a lower BHP than the flowing value, because the flowing value carried friction-gradient content that the mass budget didn't.

The old lumped-tank formulation hid this because it didn't cross-check mass at the transition — it just propagated BHP and rebuilt mG[k] at the end of each step.

Does it affect PBU analysis? No, except at sub-minute time scales.

- At 1 min: recovery might read slightly negative
- At 10 min onward: curve is physically correct
- Final pressures at 6 hr, 1 day, 7 day, 90 day: all consistent with diffusivity theory to within 1-2 psi

Workaround for users: Discard the first ~30 seconds of shut-in data when reading PBU plots. The transient is visually compressed to near-zero width on any normal log-time or hour-scale axis.

Proper fix (future work): Rewrite the flowing-state cascade to do a single consistent upward nodal integration from BHP to WHP using proper friction and holdup models, with no separate P[0] override. This would also improve pressure accuracy in the upper tubing during flowing, but touches the core flowing-state physics and should be validated against the 44-check physics test suite separately.

Other preserved limitations (unchanged from v3.3)

- **Papay Z-factor density dip near Ppr ~ 3.5** — known artifact of the Papay correlation itself; affects wells at $P_r \approx 4500\text{--}5500$ psi with sg 0.65. Would be resolved by switching to DAK 1975 or Peng-Robinson.
- **Regime classification uses instantaneous HL** — carries O(dt) truncation; see Issue 1 analysis from the earlier session.
- **No thermal wellbore storage** — gas temperature during shut-in is held at the production-time geothermal profile rather than cooling due to decompression. Second-order for most wells.
- **No gas segregation through liquid columns** — for high-WGR wells (>200 bbl/MMscf) with significant liquid fill, real PBU can show a mild “humped” response as gas rises through the liquid during shut-in. Not captured; would require an explicit drift-flux term in the shut-in column solver.

Verification summary (from probe runs)

Test	Result
Hydrostatic self-consistency	0.0002 psi max error across 20 cells
Mass conservation (90 days)	Column Δm + Reservoir Δm = net inflow from Pr boundary (infinite-acting, correct)
IPR sanity (flowing)	6.2 MMscfd at 2990 BHP vs predicted 2992 — match to 2 psi
Time-scale theory	80% recovery at 7 days matches $r_e^2/(4\eta) = 11.3$ days prediction
First-second transient	~300-500 psi BHP dip — documented as known limitation